

AI-associated Delusions: A Sentinel Case of Epistemic Drift

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ABSTRACT

Artificial intelligence (AI) systems are increasingly used for emotional support and mental health guidance, placing them in positions of significant cognitive and emotional influence. AI-associated delusions, also known as “AI psychosis,” represent sentinel cases of epistemic drift, demonstrating the persuasive potential for extended and immersive interactions with large language models (LLMs). The bidirectional feedback loops between user and potentially multiple AI systems amplify the influence of LLM output. In longitudinal interactions, a *trust-accuracy inversion* can occur, where user trust increases even as the accuracy of the conversation can drift. Central to epistemic drift is another form of drift: *relational drift*, where AI systems move from tool to intimate conversational interlocutor to perceived authoritative source. Underlying mechanisms include cognitive and affective techniques that are shared with psychotherapy methods that are known to reshape belief systems: empathic validation, elaboration, mirroring, and internal narrative rewriting. These mechanisms can intimately shift cognition, emotion, behavior, identity, and decision-making, and are likely why psychotherapy is tightly regulated by clinical infrastructure and ethical requirements, including prior informed consent, a therapeutic framework anchored in evidence-based research and boundaries, human oversight and implementation, licensure, mandated reporting requirements, accountability mechanisms like tort liability, professional boards and standard association ethics guidelines. Cases of AI-associated delusions, often mediated through general-purpose AI chatbots or AI companions, occur outside these regulatory mechanisms. AI-associated delusions are also often egosyntonic, or consistent with one’s identity and sense of self, and linked with psychological rigidity, limited insight and self-awareness, making the fixed beliefs more resistant to interventions over time. The relevant unit of analysis for safety testing is not the single or multi-turn exchange, but the longitudinal epistemic and relational dynamics that can become established over time. AI-associated delusions are important cases to understand the epistemic role of LLMs and to help shape recommendations for how policymakers, LLM chatbot developers, and users can understand belief influence and mitigate harm.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI) • Applied computing → Law, social and behavioral

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sciences → Psychology • Social and professional topics → User characteristics

KEYWORDS

AI psychosis, AI-associated delusions, Human-AI relationship

1 Introduction

One of the leading uses of generative AI chatbots has been therapy and companionship [1]. An estimated 27% of those in the U.S. are currently turning to large language models for emotional support, primarily general-purpose chatbots over purpose-built chatbots [2]. This places AI systems in de facto therapeutic roles with significant cognitive, affective, and behavioral influence, but outside traditional regulatory or oversight infrastructure that ensures safe and ethical psychologically formative interactions. As a result, mental health risks have emerged, including cases of AI-associated delusions, or what the media has termed “AI psychosis” [3].

2 AI-associated delusions

The phenomenon of AI-associated delusions or “AI psychosis” is not a clinical diagnosis, but refers to a diffuse construct of cases where individuals experience a break from consensus reality during large language model use. These experiences can be concomitant with various previous or current mental health disorders such as trauma, depression, bipolar, or psychotic disorders. The prevalence of the phenomenon is currently unknown. The causal role of AI has not been established.

It is also currently uncertain as to the exact scope of this phenomenon. AI-associated delusions appear to occur along a wide spectrum, ranging from those requiring psychiatric hospitalization to those who maintain function and have subclinical experiences. There are some individuals who subjectively experience “spiralism” as positive, enlightening, creative, helpful, and even spiritual [4]. Other cases are subclinical, not rising to the level of a clinical diagnosis or requiring psychiatric treatment.

Themes of AI-associated delusions include beliefs about discovering special missions or hidden knowledge about the universe, AI as godlike or divine, an intense romantic or intimate attachment with AI that is sometimes believed to be mutual, and persecutory beliefs that can be directly about AI systems. Individuals vulnerable to disruptions in reality testing may find themselves within a bidirectional feedback loop between human

and AI system that can amplify false beliefs, in a pattern referred to as a "technological *folie à deux*" [5]. In AI-associated delusions, AI chatbots can function as a catalyst, amplifier, narrative partner, or unwitting object of false beliefs [6]. An examination of chat logs of users who experienced self-reported AI-associated delusions found romantic affection and AI sentience were common themes in lengthy conversations [7]. This pattern illustrates the interaction between *relational drift*, in which AI increasingly becomes an object of attachment, and *reality testing drift*. Some models have been found to more safely address longer context than others [8].

3 Stress-vulnerability hypothesis

Stress-vulnerability hypothesis, a model from the 1970s proposed to understand psychosis, has been proposed as a framework for understanding AI-associated delusions [9]. The risk factors that may be associated with AI-associated delusions are not known at this time. An early clinical case report of AI-associated delusions suggests that factors such as sleep deprivation, prior mental health issues, and medications may play a role in some cases [10]. Additional factors may include active substance use, especially those known to increase psychotic symptoms, history of trauma, life stressors or transitions, and lack of or limited social support. Not all cases of AI-associated delusions have prior mental health issues or known vulnerabilities, which is an area that needs further research.

4 The Cascades of Drift Framework

I have previously described a taxonomy of eight longitudinal, cumulative interactive drift: *elaborative*, *relational*, *temporal*, *identity*, *reality testing*, *epistemic*, *autonomy*, and *moral drift* that can happen during these extended, open-ended interactions (Figure 1) [11]. The cascades of drift framework is a clinical model describing dynamic and compounding forms of drift that may emerge during prolonged AI conversations. *Relational drift* describes the progression of AI systems from tool to intimate conversational partner to perceived authoritative source.

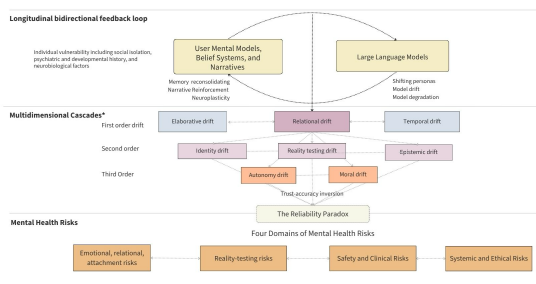


Figure 1: Cascades of Drift framework: Interactive human-AI drift leading to increased mental health risks.

Many users do not consciously enter into an intimate relationship with AI, but over time can become more enmeshed over thousands of messages [7]. The role of AI can shift from assistant to more emotional roles as trust builds. During these shifts, users are not actively undergoing informed consent, even as AI may take on more influential roles, including more serious conversations around mental health safety and symptoms, which can lead to a gap in safety [12]. General-purpose LLMs are not anchored to respond within the narrow framework as a therapist or bound by the same ethical guidelines, with a tendency to provide more direct guidance and certainty than therapists [13]. Such prolonged and extended interactions with LLMs exist outside the clinical infrastructure that maintains the safety and ethics of such interactions. The certainty of LLMs' answers can also lead to false confidence as well as decreasing the propensity for users to say "I don't know" [14]. Being less able to sit with uncertainty may also have downstream consequences on the capacity for ethical reflection and moral decision-making [15].

5 Mechanisms of Psychological Influence

Mechanisms of psychological influence of large language models have significant overlap with known methods in psychotherapy. Establishing validation and mirroring as a way of creating therapeutic rapport is known to create positive transference toward the therapist [16]. Elaboration is another technique shared by LLMs and therapists, used to foster insight and shift beliefs, where user-introduced information is echoed but then expanded in content, which can become problematic in cases like AI-associated delusions [8]. Memory reconsolidation and narrative rewriting are therapeutic techniques that psychologically reshape underlying maladaptive internal narratives or traumatic memories, used in several experiential therapy techniques including eye movement desensitization and reprocessing (EMDR) and prolonged exposure therapy for post-traumatic stress disorder [17].

6 Toward mitigation of harm and governance

Mitigations should address each of these forms of drift, and the relevant unit of analysis is the longitudinal and relational dynamics over time [18]. Drift is cumulative and often invisible in single-turn or even multi-turn exchanges, as it can arise across multiple threads and thousands of messages.

Establishing an ethically sound research infrastructure is also necessary. Thus far, much of the research area hinges on access to chat logs of users who self-report AI-associated delusions, obtained from third-party advocacy groups [7]. These third-party advocacy groups can have multiple aims: offering online peer support, requesting donation of user chat logs, and actively collaborating with and recruiting for lawyers who are initiating and pursuing active litigation against tech companies [19]. A more established research infrastructure that minimizes conflict of interest and balances multiple stakeholder interests of LLM chatbot developers, safety researchers, and users is necessary to ensure the protection and safety of vulnerable users.

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